Low-Power Configurable Multi-Clock System

Full RTL-to-GDSII ASIC Design, Implementation & Sign-Off Verification Report

---

Author: Ahmed Hossam eldin Osman
Project Top: `System_Top`
Technology: TSMC CL013G RVT (130 nm, 7-Metal Layer)
Standard Cell Library: `scmetro_tsmc_cl013g_rvt`
Operating Voltage: 1.08 V (SS), 1.20 V (TT), 1.32 V (FF)
EDA Toolchain: Synopsys (DC, DFTC, Formality, PrimeTime, SpyGlass)
Backend Toolchain: Cadence First Encounter, Mentor QuestaSim

Abstract

This technical report documents the complete design, ASIC implementation, and sign-off verification of a high-performance, low-power, configurable multi-clock digital subsystem. Designed in Verilog and SystemVerilog, the architecture integrates a full-duplex UART communication controller, a dual-clock asynchronous FIFO with Gray-coded pointer synchronization, a 2-stage multi-bit CDC bus synchronizer, a central command and arbitration finite state machine controller (`Sys_Ctrl`), a 16×8-bit register file, and an 8-bit arithmetic logic unit (ALU) supporting 15 distinct operations with active-low clock gating. The design bridges two independent asynchronous clock domains: a 100 MHz reference clock domain (`REF_CLK`) and a 3.6864 MHz UART clock domain (`UART_CLK`) alongside runtime-configurable divided clocks (`RX_Clock` and `TX_Clock`).

The implementation and verification flow encompasses RTL linting and structural/functional CDC/RDC analysis using Synopsys SpyGlass (0 violations across all 39 crossings), multi-corner logic synthesis in Synopsys Design Compiler, scan insertion (4 scan chains, 99.61% stuck-at test coverage) in DFT Compiler, formal equivalence verification in Synopsys Formality across post-synthesis, post-DFT, and post-PnR stages (100% pass rate, 362 compare points), full physical design (PnR) in Cadence SoC Encounter with complete clock tree synthesis (CTS), NanoRoute 6-metal routing, filler and dummy metal fill insertion, and zero DRC/LVS/Antenna/Density violations. Sign-off is validated by gate-level simulation (GLS) in QuestaSim with full back-annotated physical delays (SDF-max, 61/61 test vectors passing with zero timing errors) and dynamic time-based power estimation in Synopsys PrimeTime PX with GLS VCD activity and extracted DSPF parasitics (0.846 mW / 845.9 μW total average power).

October 13, 2026

Contents

1 Introduction and Specifications **2**
1.1 UART Interface & Framing Protocol . . . 2
1.2 Command Architecture . . . 2

2 System Architecture and Submodule Design **3**
2.1 Detailed Submodule Breakdown . . . 3
2.2 Project Repository . . . 4

3 Multi-Clock Architecture, CDC/RDC, and Static Verification **6**
3.1 Clock and Reset Domain Partitioning . . . 6
3.2 Clock-Domain Crossing (CDC) Implementations . . . 6
3.3 Reset-Domain Crossing (RDC) Architecture . . . 7
3.4 SpyGlass CDC & RDC Static Sign-Off . . . 7

4 Logic Synthesis and Timing Constraints **7**
4.1 Design Constraints Specification . . . 8
4.2 Synthesis Results and Cell Area Distribution . . . 8
4.3 Synthesis Timing and Power Summary . . . 8

5 Design-for-Test (DFT) and Scan Insertion **10**
5.1 Scan Architecture and Test Protocol . . . 10
5.2 ATPG Design Rule Checking & Fault Coverage . . . 11

6 Functional Simulation and Verification **11**
6.1 Verification Summary and Coverage . . . 11
6.2 Representative Command Walkthroughs . . . 11
6.3 Multi-Clock Synchronization and Latency Traceability . . . 13
6.4 FIFO-Full Back-Pressure Verification . . . 14

7 Physical Design and Implementation (Place and Route) **14**
7.1 Physical Design Flow Overview . . . 14
7.2 Clock Tree Synthesis (CTS) Analysis . . . 15
7.3 Post-Route Timing Slacks and Design Rule Violations . . . 16
7.4 Physical Verification (DRC, LVS, Antenna, Density) . . . 16

8 Formal Equivalence Checking (Formality) **16**
8.1 Equivalence Checking Strategy . . . 16
8.2 Verification Results Across All Stages . . . 17

9 Gate-Level Simulation (GLS) **17**
9.1 Simulation Configuration & Testbench Adaptation . . . 17
9.2 Timing Check Results & Functional Pass Rate . . . 18
9.3 Waveform Snippet Analysis . . . 18

10 Sign-Off Power Analysis (PrimeTime PX) **18**
10.1 Power Analysis Methodology . . . 20
10.2 Power Analysis Results . . . 20
10.3 Cross-Stage Power Comparison . . . 20

11 Conclusion and Sign-Off Matrix **22**
11.1 Key Technical Achievements . . . 22

1 Introduction and Specifications

The **Low-Power Configurable Multi-Clock System** is an integrated system-on-chip digital subsystem tailored for embedded host communication, on-chip register configuration, and high-throughput data processing under strict power and multi-clock constraints. The design processes incoming serial commands over a universal asynchronous receiver-transmitter (UART) interface, extracts target register addresses or operand vectors, computes mathematical and logical results via a dedicated ALU, and returns formatted response packets to the host.

Table 1: System Specifications and Key Design Metrics

Parameter	**Design Value / Implementation Details**
Top-Level Module	`System_Top`
Design Technology	TSMC CL013G RVT (130 nm generic process, 7 metal layers)
Primary Functional Clocks	`REF_CLK` (100 MHz, 10 ns period), `UART_CLK` (3.6864 MHz, 271.27 ns period)
Generated / Gated Clocks	`ALU_Clock` (100 MHz gated), `RX_Clock` (configurable), `TX_Clock` (115.2 kHz)
Test Clock	`scan_clk` (10 MHz, 100 ns period)
Data Width	8-bit operands and bus width; 16-bit ALU output
Register File	16 registers × 8-bit; REG0/REG1 (ALU), REG2 (UART cfg), REG3 (TX div)
Asynchronous FIFO	8 entries depth × 8-bit width, dual-clock Gray-coded pointers
Command Protocols	4 frame types: `0xAA` (Write), `0xBB` (Read), `0xCC` (ALU w/ op), `0xDD` (ALU w/o op)
Pre-DFT Area / Gates	19,734.44 μm^2 cell area, 1,828 cells (1,431 combinational, 366 sequential)
Post-DFT Area / Overlap	22,006.64 μm^2 cell area (+11.51% scan overhead), 4 scan chains, 99.61% coverage
Post-PnR Die Dimensions	240.47 μm × 160.47 μm (Core: 228.32 μm × 148.32 μm), 75.44% core utilization (100% post-fillers)
Post-Route Timing Slacks	Setup WNS = +0.364 ns, Hold WNS = +0.074 ns, 0 DRVs (Max Cap/Tran/-Fanout)
Physical Verification	0 DRC violations, 0 LVS/Connectivity errors, 0 Antenna, 0 Density errors
Formal Equivalence	100% Verified (Formality: post-syn 361 pts, post-DFT 362 pts, post-PnR 362 pts)
Gate-Level Simulation	61/61 vectors PASSED with SDF-max annotation, 0 timing check violations
Sign-Off Dynamic Power	0.846 mW (845.9 μW) total dynamic & leakage power (PrimeTime PX time-based)

1.1 UART Interface & Framing Protocol

The external link utilizes a full-duplex UART framing standard operating by default at 115,200 baud. The frame consists of 1 start bit (logic 0), 8 data bits (transmitted LSB first), 1 optional parity bit (software selectable between even and odd), and 1 stop bit (logic 1). At the default prescale factor of 32, each bit interval spans 32 clock cycles of `UART_CLK` (32×271.267 ns = 8.6805 μs per bit, matching 115,200 bps). A full 11-bit framed transmission requires 95.486 μs.

1.2 Command Architecture

The system executes four primary command formats received sequentially on `UART_RX_IN`:

1. **Register File Write (`0xAA`):** Followed by an 8-bit target address byte (`0x00` to `0x0F`) and an 8-bit data payload byte. The payload is written into the selected register upon verification. No response frame is emitted.

2. **Register File Read (`0xBB`):** Followed by an 8-bit target address byte. The requested register value is accessed, transferred through the asynchronous FIFO, and serialized back to the host on `UART_TX_O` as a 1-byte response.

3. **ALU Operation with Operands (`0xCC`):** Followed by 3 bytes: Operand A, Operand B, and ALU Function code (0 to 14). Operands are written to REG0 and REG1, the ALU executes the specified operation with clock gating asserted, and the 16-bit result is split into two consecutive response bytes (LSB followed by MSB) transmitted back to the host.

4. **ALU Operation without Operands (`0xDD`):** Followed by a single function code byte. The ALU reads the current contents of REG0 and REG1, computes the result, and returns 2 response bytes.

2 System Architecture and Submodule Design

The system architecture partitions operations across two asynchronous domains and ten modular submodules to maximize modularity and enable aggressive low-power clock gating. Figure 1 presents the top-level block diagram.

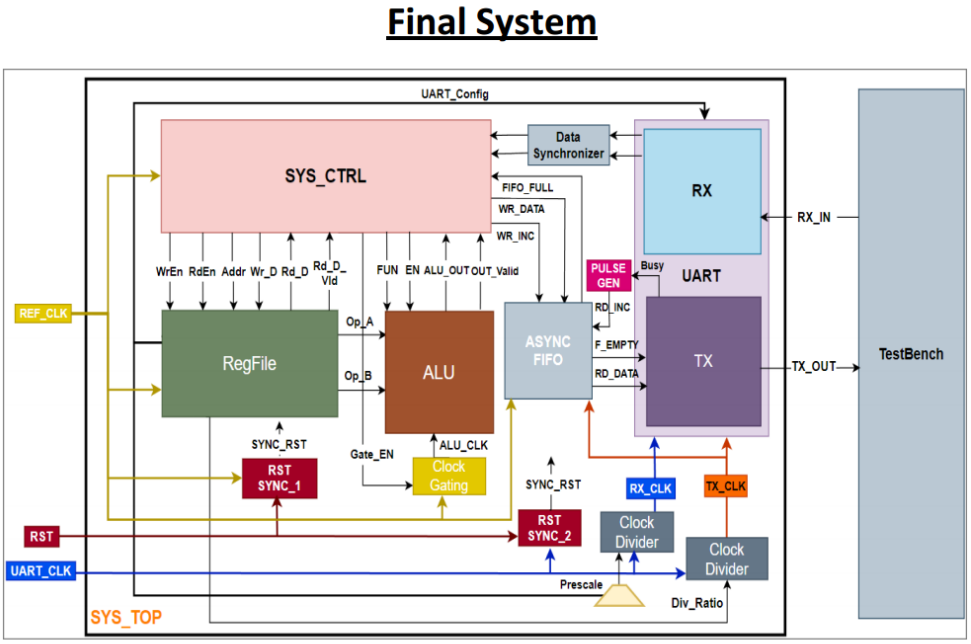


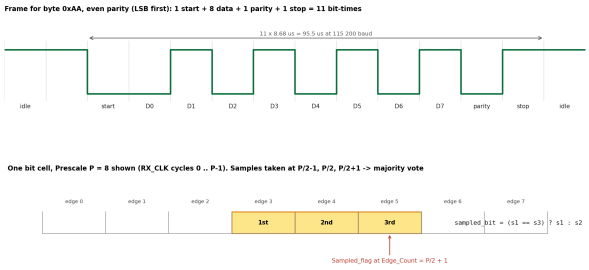
Figure 1: System block diagram showing inter-module communication, clock domains, and control buses.

2.1 Detailed Submodule Breakdown

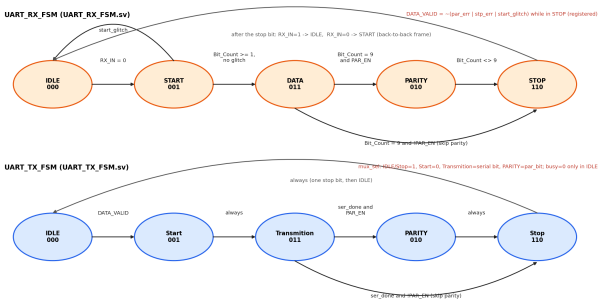
- **UART Subsystem (`u_UART`):** Comprises `UART_RX` and `UART_TX_TOP`. The receiver incorporates seven internal units: `Start_Checker`, `Sampler`, `Edge_Bit_Counter`, `Deserializer`, `Parity_Checker`, `Stop_Checker`, and `UART_RX_FSM`. The sampler implements a 3-sample majority voter (samples taken at $P/2-1$, $P/2$, and $P/2+1$, where P is the prescale division factor) to filter glitches and sample jitter (Figure 2a). The transmitter comprises `UART_Serializer`, `UART_Parity`, `UART_TX_FSM`, and a 4-to-1 output multiplexer selecting between start, data, parity, and stop bits (Figure 2b).

- **CDC Bus Synchronizer (`u_BUS_SYNC`):** Bridges parallel command bytes from the UART domain to the reference domain. The multi-bit data lines are held stable while a 2-stage flip-flop synchronizer synchronizes the accompanying valid strobe. The synchronized strobe triggers a local pulse generator in `REF_CLK`, enabling multi-cycle hold without multi-bit metastability skew.

- **Dual-Clock Asynchronous FIFO (`u_FIFO`):** Bridges the response data path from the 100 MHz `REF_CLK` domain to the slow 115.2 kHz `TX_Clock` domain. Pointer comparison is conducted in Gray code across 2-stage synchronizers (`DF_SYNC`), preventing transient multi-bit crossing artifacts.
- **System Controller (`u_SysCtrl`):** A central finite state machine operating on `REF_CLK`. It sequences through command capture, register file decoding, ALU operand loading, execution gating, response push, and back-pressure stall handling (Figure 3).
- **Register File (`u_RegFile`):** Houses 16 addressable 8-bit registers (Table 2). Registers 0 and 1 automatically feed ALU operands A and B. Register 2 stores the UART configuration (Prescale ratio in bits [7:2], Parity Enable in bit 0, and Parity Type in bit 1). Register 3 stores the TX clock division ratio (default 32). Registers 4 to 15 serve as general-purpose scratchpad storage.
- **Arithmetic Logic Unit (`u_ALU`):** Supports 15 operations (Table 3). Output width is 16 bits to preserve the full dynamic range of multiplication.
- **Clock Gating Cell (`u_CLK_GATE`):** Suppresses `ALU_Clock` toggling when the ALU is idle. Controlled by `u_pulse_gen`, which generates an enable pulse asserted for exactly the required execution cycle.
- **Reset Synchronizers (`u_REF_RST_Sync`, `u_UART_RST_Sync`):** Provide asynchronous assertion and synchronous de-assertion per clock domain, preventing recovery/removal timing violations during chip startup.



(a) UART frame format and mid-bit majority sampling.



(b) UART RX and TX finite state machine diagrams.

Figure 2: UART interface physical signaling and FSM architectures.

Table 2: Register File Address Mapping and Defaults

Address	Register Name	Reset Value	Access	Hardware Purpose / Connectivity
0x00	REG0	0x00	Read / Write	ALU Operand A
0x01	REG1	0x00	Read / Write	ALU Operand B
0x02	REG2	0x81	Read / Write	UART Config: Prescale [7:2] (32), Parity Typ [1], Parity En [0]
0x03	REG3	0x20	Read / Write	TX Clock Division Ratio (default 32)
0x04--0x0F	REG4--15	0x00	Read / Write	General-Purpose Scratchpad Registers

2.2 Project Repository

View the full project repo and RTL design files on GitHub:
https://github.com/ahmedhossam9128/Low-Power-Configurable-Multi-Clock-System

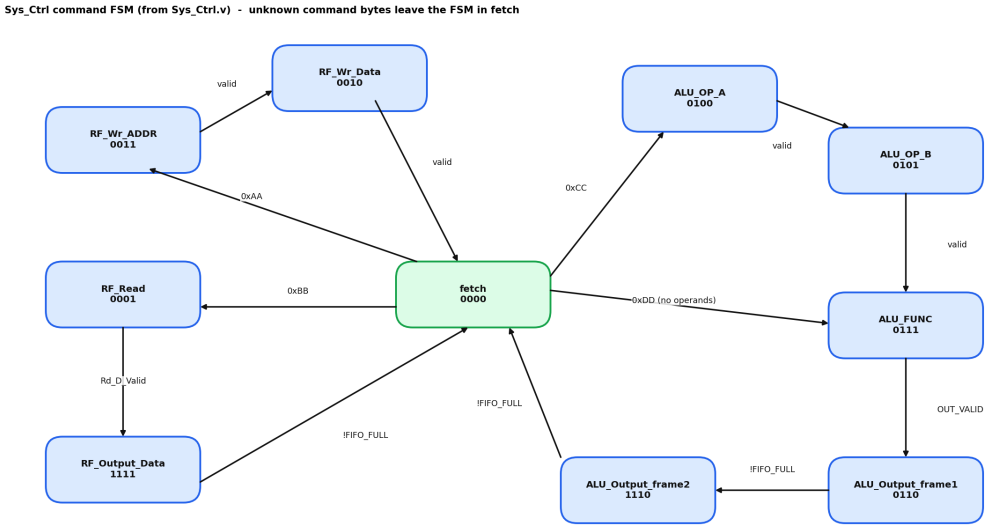


Figure 3: System Controller (`Sys_Ctrl`) state transition diagram with 4-bit state encodings.

Table 3: ALU Function Encodings and Operations

Function Code	Operation Name	Mathematical Definition	Output Format / Bit Width
0x0	Addition (ADD)	$A + B$	16-bit sum (handles carry)
0x1	Subtraction (SUB)	$A - B$	16-bit difference (handles borrow)
0x2	Multiplication (MUL)	$A \times B$	16-bit full product
0x3	Division (DIV)	$A \div B$ (quotient)	16-bit quotient; 0 if $B = 0$
0x4	Bitwise AND	$A \,\&\, B$	16-bit zero-extended
0x5	Bitwise OR	$A \mid B$	16-bit zero-extended
0x6	Bitwise NAND	$\sim (A \,\&\, B)$	16-bit ones-extended upper byte
0x7	Bitwise NOR	$\sim (A \mid B)$	16-bit ones-extended upper byte
0x8	Bitwise XOR	$A \hat{} B$	16-bit zero-extended
0x9	Bitwise XNOR	$\sim (A \hat{} B)$	16-bit ones-extended upper byte
0xA	Equivalence (EQ)	$(A == B)\ ?\ 1 : 0$	16-bit boolean flag (bit 0)
0xB	Greater Than (GT)	$(A > B)\ ?\ 2 : 0$	16-bit boolean flag (bit 1)
0xC	Less Than (LT)	$(A < B)\ ?\ 3 : 0$	16-bit boolean flag (bits [1:0])
0xD	Shift Right (SHR)	$A \gg 1$	16-bit zero-extended
0xE	Shift Left (SHL)	$A \ll 1$	16-bit zero-extended (bit 8 carry)

3 Multi-Clock Architecture, CDC/RDC, and Static Verification

3.1 Clock and Reset Domain Partitioning

The subsystem features two primary asynchronous clock trees and three derived clock networks, summarized in Table 4. The full domain mapping is illustrated in Figure 4.

Table 4: System Clock Domains and Frequency Specifications

Clock Net	Nominal Freq.	Period	Source Type	Driven Submodules / Purpose
`REF_CLK`	100.0 MHz	10.000 ns	Primary Port	`Sys_Ctrl`, `Reg_File`, FIFO Write, Bus Sync dest
`ALU_Clock`	100.0 MHz	10.000 ns	Gated	`u_ALU` execution datapath
`UART_CLK`	3.6864 MHz	271.267 ns	Primary Port	UART Rx sampling, Clock dividers
`RX_Clock`	Configurable	Configurable	Generated	UART Rx bit tracking (Prescale 32/16/8)
`TX_Clock`	115.2 kHz	8680.533 ns	Generated	`UART_TX_TOP`, FIFO Read side
`scan_clk`	10.0 MHz	100.000 ns	Test Port	DFT Scan shifting/capture (test mode only)

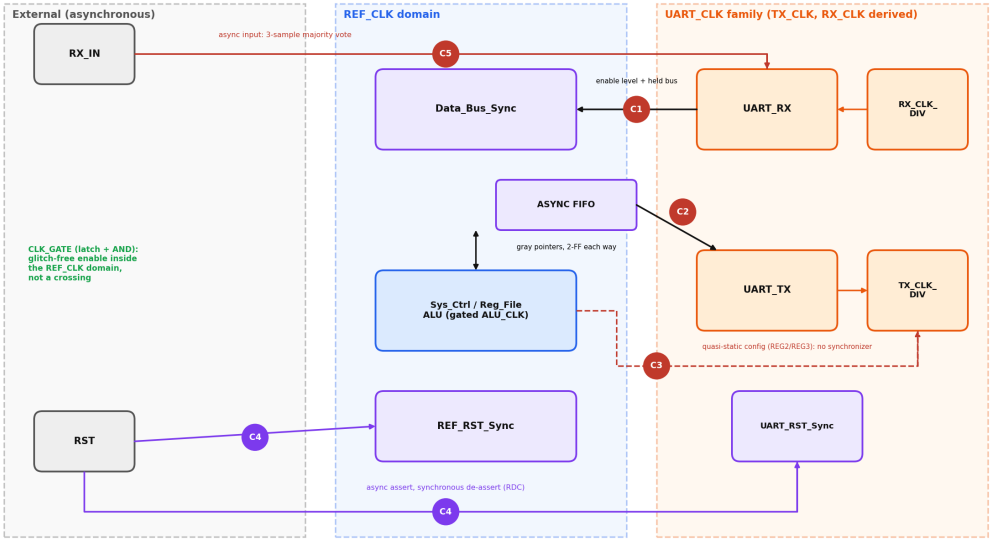


Figure 4: Clock and reset domain topology depicting the five asynchronous boundaries.

3.2 Clock-Domain Crossing (CDC) Implementations

Crossings between asynchronous domains are handled using strictly validated architectural primitives:

1. **Control Strobe & Bus Synchronizer (`UART_CLK` → `REF_CLK`):** The UART byte deserializer presents `RX_P_DATA` alongside `RX_D_VALID`. Rather than passing data bits through parallel flip-flops (which would suffer from bus skew and re-convergence hazards), `RX_D_VALID` is passed through a two-stage flip-flop synchronizer in `u_BUS_SYNC`. A trailing pulse generator produces a single-cycle pulse in `REF_CLK` that latches the multi-bit parallel bus, which is guaranteed to be stable across several reference cycles.

2. **Dual-Clock FIFO Pointer Crossing (`REF_CLK` ↔ `TX_Clock`):** The FIFO write pointer (`w_ptr`) is converted from binary to Gray code before crossing into the `TX_Clock` domain through a 2-stage synchronizer (`DF_SYNC`). Gray code ensures that exactly one bit changes per count transition, eliminating multi-bit glitch states. An identical Gray synchronizer transfers the read pointer (`r_ptr`) back to `REF_CLK` for full condition evaluation (Figure 5).

3. **Quasi-Static Configuration Registers (`Reg_File` → UART / Dividers):** Register bits in `REG2` and `REG3` change only upon host command write and remain static during normal transmission. These paths are formally constrained as quasi-static.

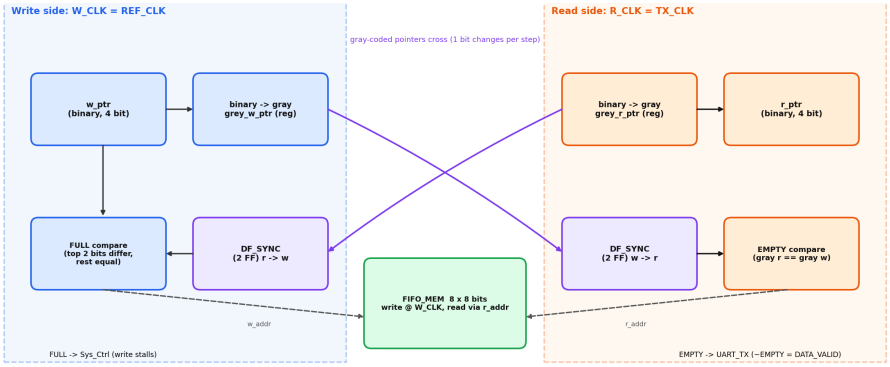


Figure 5: Asynchronous FIFO architecture showing Gray-coded read/write pointer synchronizers.

3.3 Reset-Domain Crossing (RDC) Architecture

To prevent race conditions during reset release, the global reset `RST_N` is asynchronously asserted and synchronously de-asserted using dedicated two-stage reset synchronizers (`u_REF_RST_Sync` and `u_UART_RST_Sync`). Each synchronizer cascades two flip-flops with their data inputs tied to logic 1 and active-low asynchronous resets tied to `RST_N`. This guarantees that the internal synchronous reset drops synchronously on a clock rising edge, preserving recovery and removal margins.

3.4 SpyGlass CDC & RDC Static Sign-Off

Static verification was executed using Synopsys SpyGlass against the industry-standard `rtl_handoff` methodology. Design constraints were authored in `System_Lint/system.sgdc`:

```
current_design System_Top
sdc_data -file ./../DFT/sdc/System_Top.sdc
reset -name RST -value 0 -async

quasi_static -name u_RegFile.REG2
quasi_static -name u_RegFile.REG3
quasi_static -name u_RX_CLK_DIV_MUX.RX_CLK_DIV_RATIO
quasi_static -name u_SysCtrl.CLK_DIV_EN

reset_synchronizer -name u_REF_RST_Sync.sync_RST
reset_synchronizer -name u_UART_RST_Sync.sync_RST
sync_cell -name Data_Bus_Sync
sync_cell -name DF_SYNC

cdc_attribute -unrelated {System_Top.u_RegFile.REG2 System_Top.u_RX_CLK_DIV.o_div_clk}
cdc_attribute -unrelated {System_Top.u_RegFile.REG3 System_Top.u_TX_CLK_DIV.o_div_clk}
```

Listing 1: System SGDC Constraints for SpyGlass CDC and RDC

All reported messages were audited and approved with zero functional waivers or open defects.

4 Logic Synthesis and Timing Constraints

Synthesis was performed in Synopsys Design Compiler (vO-2018.06-SP1) mapping the design to the TSMC CL013G RVT 130 nm standard cell library (`scmetro_tsmc_cl013g_rvt_ss_1p08v_125c.db`).

Table 5: SpyGlass CDC, RDC, and Linting Sign-Off Summary

Goal / Policy	Rule ID	Violations	Waived	Sign-Off Verification Status
RTL Lint (`lint_rtl`)	All Rules	0	0	Fully Clean; 0 errors, 0 warnings
Structural CDC	`Ac_unsync01`	0	0	0 Unsynchronized scalar crossings (11 sync)
	`Ac_unsync02`	0	0	0 Unsynchronized vector crossings (11 sync)
	`Quasi-static`	0	0	17 Verified quasi-static crossings
Functional CDC	`Ac_conv01--04`	0	0	0 Sequential/combinational convergence hazards
	`Ac_glitch03`	0	0	0 Glitches on control crossing paths
	`Ac_cdc08`	0	0	0 Race conditions on crossing control buses
Reset Domain	`Ar_unsync01`	0	0	0 Unsynchronized resets; 2 sync verified
	`Ar_syncdeassert01`	0	0	2 Verified synchronous reset deassertions
	`Ar_resetcross01`	0	0	0 RDC violations detected

4.1 Design Constraints Specification

Constraints were authored in `cons.tcl` and exported to `System_Top.sdc`:

- **Reference Clock:** `create_clock -name REF_CLK -period 10.0 [get_ports REF_CLK]` (100 MHz, 50% duty cycle). Clock uncertainty set to 0.2 ns setup, 0.1 ns hold.
- **UART Clock:** `create_clock -name UART_CLK -period 271.267 [get_ports UART_CLK]` (3.6864 MHz)
- **Generated Clocks:**
  - `ALU_Clock`: Derived from `u_CLK_GATE/GATED_CLK` with divide factor 1.
  - `RX_Clock`: Derived from `u_RX_CLK_DIV/o_div_clk`.
  - `TX_Clock`: Derived from `u_TX_CLK_DIV/o_div_clk` with divide factor 32.
- **Asynchronous Domain Declaration:** `set_clock_groups -asynchronous -group [get_clocks {REF_CLK ALU_Clock}] -group [get_clocks {UART_CLK RX_Clock TX_Clock}]`.
- **I/O Timing and Environmental Constraints:** Input maximum delay constrained to $0.2 \times T_{clk}$; output maximum delay constrained to $0.2 \times T_{clk}$. Drive strength modeled using `BUFX2M`; capacitive load modeled as 0.5 pF on outputs.

4.2 Synthesis Results and Cell Area Distribution

Pre-DFT synthesis optimized with `compile_ultra -no_autoungroup` yielded a total cell area of 19,734.44 μm^2 across 1,828 standard cells (1,431 combinational cells including 209 buffers/inverters, and 366 sequential cells). The hierarchical area distribution is detailed in Table 6.

4.3 Synthesis Timing and Power Summary

Timing was verified under worst-case conditions (SS 1.08 V, 125°C). All 5 path groups met timing with positive slack: Worst Setup Slack = +0.02 ns (WNS = +0.02 ns, TNS = 0.00 ns on `ALU_Clock`; +6.91 ns on `REF_Clock`; +214.16 ns on `UART_Clock`); Worst Hold Slack = +0.34 ns (on `REF_Clock`; +0.58 ns on `ALU_Clock`; +0.36 ns on `UART_Clock`). Constraint verification (`report_constraint -all_violators`) confirmed zero violated design rules or timing constraints. Vectorless power analysis estimated a total

Table 6: Hierarchical Cell Area Distribution (Design Compiler `compile_ultra`)

Hierarchy Module	**Total Area (μm^2)**	**Share (%)**	**Comb. Area**	**Seq. Area**	**Function / Contents**
`System_Top` (Top)	19,734.44	100.0%	10,444.39	9,290.05	Full multi-clock subsystem
`u_RegFile`	5,535.20	28.0%	1,988.62	3,546.57	16×8-bit register file
`u_ALU`	4,911.55	24.9%	4,471.46	440.09	15-function ALU datapath
`u_FIFO`	3,746.61	19.0%	1,268.48	2,478.13	Dual-clock FIFO and sync
`u_fifo_mem`	2,649.93	13.4%	993.13	1,656.79	8×8 dual-port memory core
`u_fifo_rd`	344.77	1.7%	137.67	207.10	Read pointer Gray encoder
`u_fifo_wr`	337.71	1.7%	130.61	207.10	Write pointer Gray encoder
`u_sync_r2w`	207.10	1.0%	0.00	207.10	2-stage read-to-write sync
`u_sync_w2r`	207.10	1.0%	0.00	207.10	2-stage write-to-read sync
`u_UART`	2,496.96	12.7%	1,141.67	1,355.29	UART Rx & Tx controllers
`u_UART_RX`	1,722.69	8.7%	917.48	805.21	Oversampled RX core
`u_UART_TX_TOP`	774.27	3.9%	374.19	400.08	Serializer & TX FSM
`u_SysCtrl`	1,437.93	7.3%	583.64	854.28	Central controller FSM
`u_TX_CLK_DIV`	708.37	3.6%	475.39	232.99	8-bit configurable divider
`u_BUS_SYNC`	395.37	2.0%	84.72	310.65	2-stage multi-bit bus sync
`u_RX_CLK_DIV`	238.87	1.2%	135.32	103.55	3-bit RX divider
`u_REF_RST_Sync`	77.66	0.4%	25.89	51.77	Reference reset synchronizer
`u_UART_RST_Sync`	61.19	0.3%	0.00	61.19	UART reset synchronizer
`u_pulse_gen`	57.66	0.3%	5.88	51.77	Single-cycle pulse generator
`u_RX_CLK_DIV_MUX`	36.48	0.2%	36.48	0.00	Divider output multiplexer
`u_CLK_GATE`	27.06	0.1%	0.00	27.06	Integrated clock gate cell

dissipation of 0.439 mW (Internal power 95.7%, Switching power 2.3%, Leakage power 2.1%). Figure 6 displays the post-synthesis gate-level schematic.

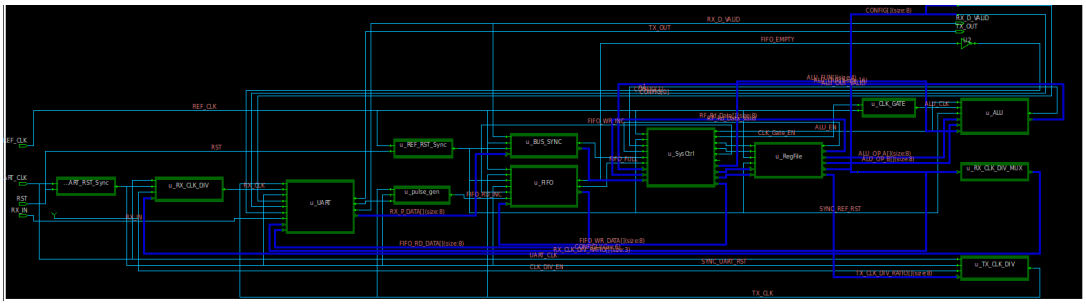


Figure 6: Gate-level schematic of `System_Top` post-synthesis in Synopsys Design Compiler.

5 Design-for-Test (DFT) and Scan Insertion

5.1 Scan Architecture and Test Protocol

Scan insertion was executed in Synopsys DFT Compiler. The functional architecture contains multiple clock domains and clock gating cells that must be made fully controllable during manufacturing test:

1. **Scan Chains:** The sequential cells were balanced across 4 independent scan chains (`NUM_OF_CHAINS = 4`), driven by scan inputs `SI[3:0]` and captured on `SO[3:0]`.
2. **Test Control Ports:** A dedicated test interface was inserted: `test_mode` (enables test configuration), `SE` (Scan Enable: shifts when high, captures when low), `scan_clk` (10 MHz test clock), and `scan_rst` (active-low test reset).
3. **Clock and Reset Muxing:** During test mode (`test_mode = 1`), functional clock lines (`REF_CLK`, `UART_CLK`, `ALU_Clock`, `RX_Clock`, `TX_Clock`) and reset lines are bypassed to `scan_clk` and `scan_rst` through dedicated 2-to-1 scan multiplexers (`MUX_2x1` instances).
4. **Clock Gate Bypass:** In test mode, the clock gating enable is forced active to ensure scan flops driven by `ALU_Clock` remain fully shiftable and capturable.

Table 7: DFT Scan Insertion Metrics and Overhead

Metric	**Pre-DFT (`compile_ultra`)**	**Post-DFT Scan Architecture**
Total Cell Area	19,734.44 μm^2	22,006.64 μm^2 (+11.51% scan area overhead)
Combinational Area	10,444.39 μm^2 (1,431 cells)	10,276.12 μm^2 (1,395 cells)
Noncombinational Area	9,290.05 μm^2	11,730.52 μm^2 (+26.27% scan flop overhead)
Total Cell Count	1,828 cells	1,800 cells
Sequential Cells	366 sequential cells	366 sequential cells (346 scan + 12 shift-reg + 1 ICG)
Total Ports / Pins	552 pins	680 pins (+128 scan/test signals)
Scan Chains	0	4 balanced chains (Chains 1–2: 90 cells, Chains 3–4: 89 cells)
Valid Scan Cells	0	346 scan-replaced flops
Clock/Reset Test Muxes	0	8 instances of `MUX_2x1` (controlled by `test_mode`)
Scan Setup / Hold Slack	N/A	+78.62 ns / +0.32 ns (on 100 ns `scan_clk`)

5.2 ATPG Design Rule Checking & Fault Coverage

Post-DFT DRC validation was executed using `dft_drc -verbose -coverage_estimate` in Synopsys DFT Compiler. The test protocol verified full controllability and observability across all 4 scan chains:

- **Scan Chain Continuity:** All 4 chains shift and capture deterministically. No clock race, hold time, or bus contention violations were detected during ATPG DRC.
- **Sequential Cell Classification:** Out of 359 sequential cells analyzed during DRC, 346 were converted into valid scan cells, 12 were identified as non-scan shift-register cells and stitched into the scan chain pipeline, and exactly 1 cell (`u_CLK_GATE/u_CLK_GATE`) had a constant 1 value (the architectural clock gating latch bypassed by `test_mode = 1`).
- **Stuck-At Fault Coverage Analysis:** The ATPG analysis added 15,604 faults to the fault list and converged in 15 optimization iterations to an exceptional **99.61% test coverage**:

$$\text{Test Coverage} = \frac{\text{Detected Faults}}{\text{Total Faults} - \text{Undetectable Faults}} = \frac{15,505}{15,604 - 38} \times 100\% = 99.61\% \quad (1)$$

  The raw fault coverage is $\frac{15,505}{15,604} \times 100\% = 99.37\%$. Out of 15,604 total faults, only 60 were ATPG-untestable and exactly 1 fault was undetected, demonstrating outstanding manufacturing defect testability.

Table 8: Synopsys DFT Compiler Uncollapsed Stuck Fault Summary Report

Fault Class	Code	Fault Count	Fault Percentage (%)
Detected	DT	15,505	99.37%
Possibly Detected	PT	0	0.00%
Undetectable	UD	38	0.24%
ATPG Untestable	AU	60	0.38%
Not Detected	ND	1	0.01%
Total Faults	–	**15,604**	**100.00%**
Test Coverage	–		**99.61%**
Fault Coverage	–		**99.37%**

6 Functional Simulation and Verification

System-level verification was performed using the self-checking SystemVerilog testbench `SIM/System_Top_tb.sv` in Mentor QuestaSim. The testbench models an asynchronous UART host framing bytes to `UART_RX_IN`, decoding response frames from `UART_TX_O`, and validating timing across clock ratios.

6.1 Verification Summary and Coverage

The test suite executed **167 comprehensive checks** covering all command sequences, register operations, 15 ALU functions, configuration matrices, and back-pressure corner cases:

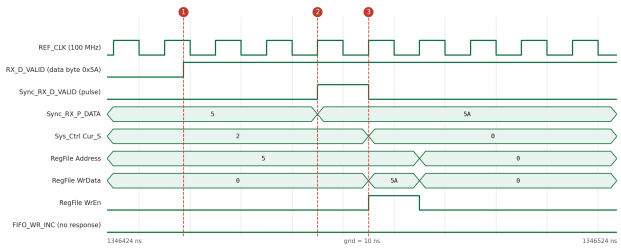
PASSED: 167 FAILED: 0 (100% Test Pass Rate)

6.2 Representative Command Walkthroughs

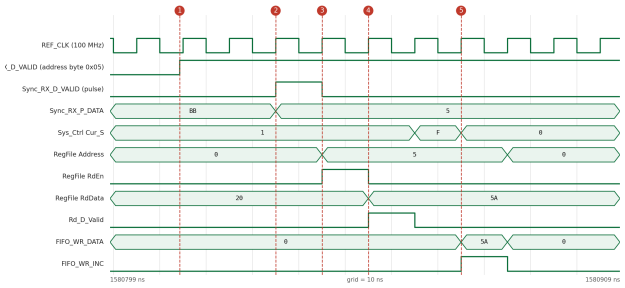
Figure 7 displays the simulation waveforms for representative write, read, and arithmetic commands:

Table 9: Functional Test Suite Verification Matrix (167 Checks)

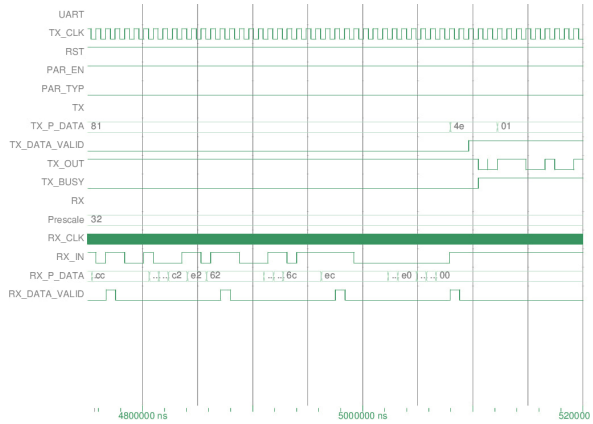
Test Phase	**Checks**	**Description / Target Coverage**
Divider Outputs	2	Verified `RX_Clock` period (271.3 ns) and `TX_Clock` period (8680.6 ns)
Config Read-Back	2	Verified applied configuration via read-back of REG2 and REG3
Register File Operations	5	Verified write silent behavior (no unexpected TX) and read at addresses 0x4, 0x7, 0xB, 0xF
ALU Operations w/ Operands	60	Full sweep: 15 ALU functions × 2 test vectors × 2 output bytes (CC command)
Register Preload	1	Confirmed REG0/REG1 write produces zero spurious TX traffic
ALU Operations w/o Operands	10	5 representative ALU functions × 2 output bytes using preloaded operands (DD command)
Mixed & Idle Checks	5	Mixed CC/DD verification, TX line verified idle upon completion
Prescale 16 and 8 Sweeps	62	31 checks each for Prescale 16 and Prescale 8 verifying configuration and register operations
FIFO Back-Pressure Phase	14	TX divider slowed to 128 (REG3 = 128); FIFO filled to 8 entries; 6 queued ALU results
Total Verification Checks	**167**	**100% Passed, 0 Failures**



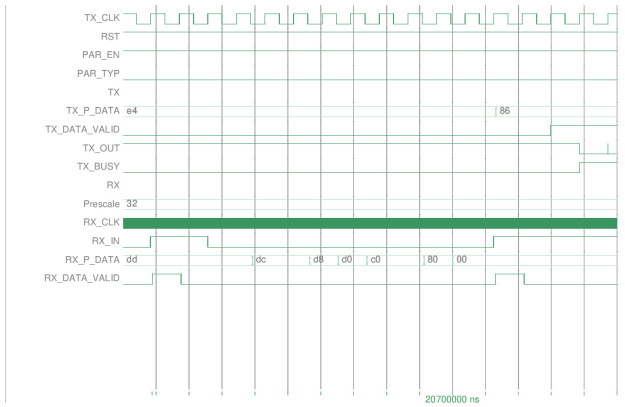
(a) Register Write (`AA 05 5A`): loads 0x5A into REG5.



(b) Register Read (`BB 05`): returns 0x5A on TX.



(c) ALU with Operands (`CC`): 16-bit result generation.



(d) ALU without Operands (`DD`): preloaded REG0/1.

Figure 7: Consolidated functional simulation waveforms for representative commands.

6.3 Multi-Clock Synchronization and Latency Traceability

Figure 8 illustrates a cycle-accurate 150 ns zoom of a `CC` command navigating both clock crossings and clock gating:

1. **Marker 1 (0 ns):** `UART_RX` raises `RX_D_VALID`.
2. **Marker 2 (+27 ns):** Synchronized pulse `Sync_RX_D_VALID` pulses for 1 `REF_CLK` cycle; function byte `0x00` (ADD) is captured.
3. **Marker 3 (+37 ns):** `SYS_CTRL` registers `ALU_EN` and `CLK_Gate_EN`; clock gate latch opens during the low clock phase.
4. **Marker 4 (+47 ns):** A single active compute edge occurs on `ALU_Clock`, producing `ALU_OUT` = `0x014E` and `OUT_VALID` = 1. The result bytes are pushed to the FIFO over the subsequent two cycles (`FIFO_WR_INC` asserted).
5. **Marker 5 (+77 ns):** Flush edge with `ALU_EN` low; clock gate shuts off, maintaining static low power until the next command.

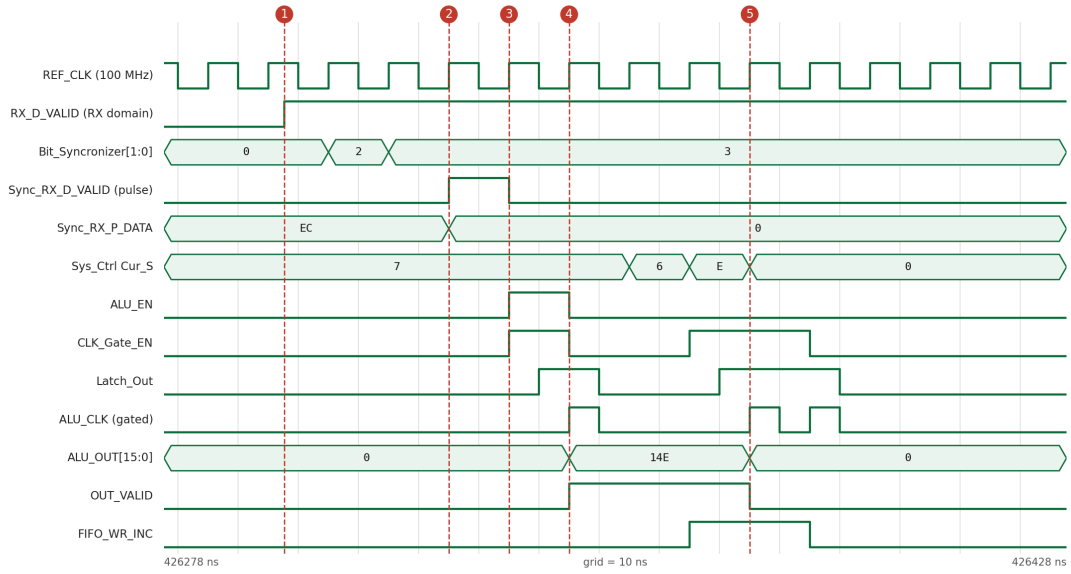
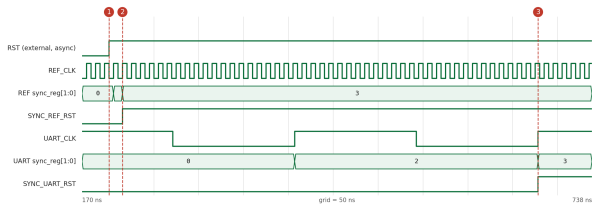
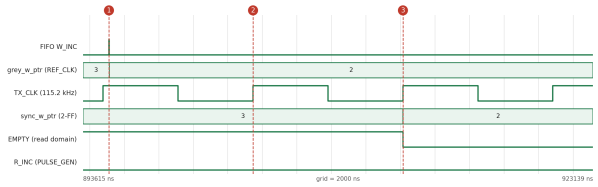


Figure 8: Cycle-accurate simulation zoom of a compute sequence across clock boundaries and clock gate.



(a) Reset release synchronization in `REF_CLK` and `UART_CLK`.



(b) FIFO Gray-code pointer crossing and 17.0 μs empty deassertion.

Figure 9: Reset deassertion and FIFO Gray pointer synchronization waveforms.

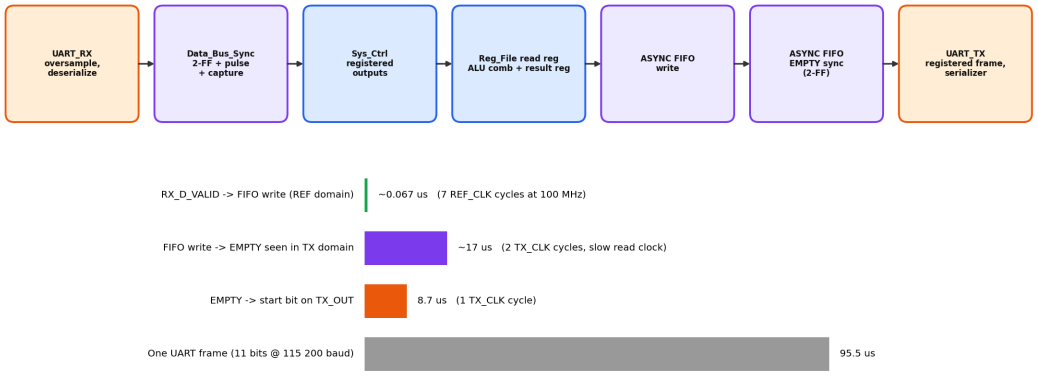


Figure 10: Command-to-response pipeline with measured latencies, drawn to scale against one UART frame (95.5 μs).

6.4 FIFO-Full Back-Pressure Verification

At the default TX divider (`REG3 = 32`), the UART transmitter drains an 8-bit frame in 95.5 μs, whereas a command byte is received in 104.2 μs. As a result, under default conditions the 8-entry FIFO never fills, leaving the `SYS_CTRL` back-pressure stall path unexercised.

To verify stall recovery, `REG3` was reprogrammed to 128, inverting the drain-to-fill ratio. Six consecutive ALU operations (generating 12 response bytes) were pushed into the 8-entry FIFO. The FIFO successfully asserted `FIFO_FULL`, stalling the controller for **20,772 `REF_CLK` cycles**. Once the transmitter drained entries, the controller resumed seamlessly, and all 12 response bytes were received in exact order without loss or corruption.

7 Physical Design and Implementation (Place and Route)

Physical implementation was carried out in Cadence First Encounter (v08.10-p004_1) using the TSMC 130 nm 7-metal layer process (`tsmc13fsg_7lm_tech.lef` and `tsmc13_m_macros.lef`).

7.1 Physical Design Flow Overview

The backend flow follows a structured RTL-to-GDSII methodology:

1. **Design Import & MMMC Setup:** Gate-level netlist loaded from `../DFT/netlists/System_Top.v` alongside scan DEF `../DFT/System_Top_scan.def`. Multi-Mode Multi-Corner (MMMC) views configured across 3 modes (`func_mode`, `capture_mode`, `scan_mode`) and 3 library corners (SS 1.08 V 125°C, TT 1.2 V 25°C, FF 1.32 V -40°C), establishing 6 active analysis views (`setup1-3` and `hold1-3`).

2. **Floorplanning & Power Planning:** Die dimensions established at 240.47 μm × 160.47 μm (aspect ratio ≈ 1.50) with 6.15 μm core margins. Core dimensions are 228.32 μm × 148.32 μm, containing 52 standard cell rows (2.87 μm row height, 0.41 μm site width, 556 sites per row, 33,355 total allocatable sites). Placement density progresses from 69.89% pre-CTS to 75.27% post-CTS (due to clock buffer/inverter tree insertion) and reaches **75.44%** post-routing (hold buffers and timing optimization). Following filler cell insertion, core row utilization reaches **100%**. Module guides placed for `u_SysCtrl` and `u_ALU`. Power rings and stripes routed on Metal 5 and Metal 6; standard cell rails routed on Metal 1.

3. **Placement & Tie Cell Insertion:** Standard cells placed with `placeDesign -inPlaceOpt -prePlaceOpt`. Tie-high (`TIEHIM`) and tie-low (`TIELOM`) cells inserted to tie unused inputs safely without direct rail connections.

4. **Clock Tree Synthesis (CTS):** Clock trees constructed using `clockDesign` based on `Clock.ctstch`. Skew balancing optimized for `scan_clk`, `UART_CLK`, and `REF_CLK`.

5. **Routing (NanoRoute):** Routed using NanoRoute with top routing layer constrained to Metal 6 (`setNanoRouteMode -routeTopRoutingLayer 6`). Concurrent via and wire optimization enabled.

6. **Chip Finishing:** Insertion of 2,045 physical filler cells (`FILL1M` through `FILL64M`) bringing standard cell row utilization from 75.44% to 100% continuous density (guaranteeing N/P-well continuity and power rail integrity), followed by dummy metal fills across layers 1–7 with density targets (20%–80%).

7. **Physical Verification:** Connectivity (LVS), Geometry (DRC), Process Antenna, and Metal Density verification executed and cleared.

8. **Sign-Off Exports:** GDSII stream-out, post-route Verilog netlist (`System_Top.v` and `System_Top_pg.v`), SPEF parasitics (`System_Top.spf`), and delay calculation (`System_Top.sdf`).

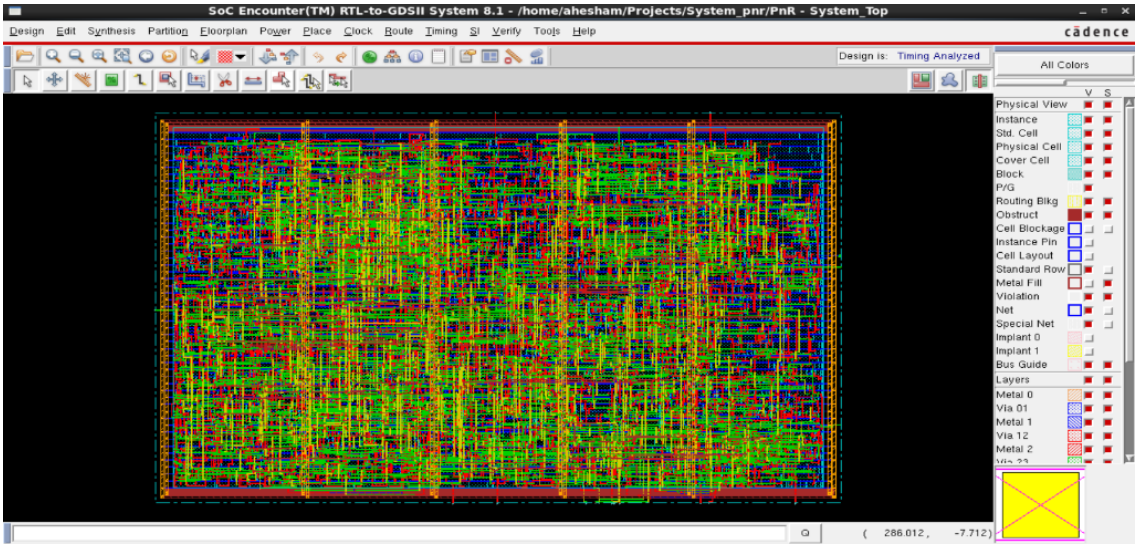


Figure 11: Cadence SoC Encounter post-routing interconnect layout (NanoRoute, Metal 1–6) prior to dummy metal fill insertion (pre-metal filling), highlighting active cell routing, clock tree synthesis (CTS) distributions, and power grid distribution without dummy fill obstruction.

7.2 Clock Tree Synthesis (CTS) Analysis

Clock tree synthesis was executed to balance insertion delay and minimize skew across the multiple domains. Target skew was set to 200 ps. The results from `PnR/clock_report/clock.report` are summarized in Table 10.

Table 10: Clock Tree Synthesis Implementation Results

Clock Tree	Sinks	Buffers	Levels	Max Edge Delay	Achieved Skew	Status / Margin
`REF_CLK`	281	47	16	1,842.1 ps	**109.3 ps**	Met (200 ps target, 45% margin)
`UART_CLK`	75	30	14	1,910.4 ps	**201.1 ps**	Met at boundary
`scan_clk`	356	92	19	1,991.5 ps	**201.1 ps**	Met test skew target

Post-CTS optimization (`optDesign -postCTS -hold`) successfully repaired hold time slacks across all corners prior to global routing.

7.3 Post-Route Timing Slacks and Design Rule Violations

Following NanoRoute and post-route timing optimization, STA was performed across all analysis views using `timeDesign -postRoute`. The setup and hold timing summaries from `PnR/timingReports/` are presented in Table 11.

Table 11: Post-Route Timing Slack Summary (Cadence Encounter)

Analysis Mode	All Paths	reg2reg	in2reg	reg2out	clkgate	Violations
Setup WNS (ns)	**+0.364**	+0.364	+8.376	+76.489	+8.445	**0**
Setup TNS (ns)	0.000	0.000	0.000	0.000	0.000	0
Total Setup Paths	1,065	1,050	363	7	1	0
Hold WNS (ns)	**+0.074**	+0.074	+0.086	+21.184	+0.416	**0**
Hold TNS (ns)	0.000	0.000	0.000	0.000	0.000	0
Total Hold Paths	1,065	1,050	363	7	1	0
Design Rule Violations (DRVs)	Max Capacitance: 0 (0 terms)			Max Transition: 0 (0 terms), Max Fanout: 0		

Both setup and hold constraints are satisfied with healthy positive margins under worst-case operating conditions.

7.4 Physical Verification (DRC, LVS, Antenna, Density)

Comprehensive sign-off physical verification checks were performed on the finished layout:

- **Connectivity (LVS Verification, `verifyConnectivity`):** Executed on all nets including power and ground. **Result: Found no problems or warnings.** Zero opens, zero shorts, zero dangling nets.
- **Geometry (DRC Verification, `verifyGeometry`):** Executed across all layers. **Result: No DRC violations were found** (Cells: 0, SameNet: 0, Wiring: 0, Antenna: 0, Short: 0, Overlap: 0).
- **Process Antenna Check (`verifyProcessAntenna`):** Checked against TSMC 130 nm antenna rules. **Result: No Violations Found.**
- **Metal Density Verification (`verifyMetalDensity`):** Evaluated across Metal 1 through Metal 7 in a 200 μm $\times$ 200 μm window with 100 μm step size. **Result: No density violations were found.** All windows satisfy the 20%–80% density window.

8 Formal Equivalence Checking (Formality)

Formal verification was performed in Synopsys Formality (vL-2016.03-SP1) to mathematically prove functional equivalence across every stage of synthesis and physical implementation.

8.1 Equivalence Checking Strategy

Formality was executed at three distinct transformation milestones:

1. **Post-Synthesis Verification (`post-syn`):** Golden RTL source files (Reference container) vs. Post-Synthesis netlist (Implementation container). SVF guidance loaded from `Synthesis/System_Top.svf`.
2. **Post-DFT Verification (`post-dft`):** Golden RTL (Reference) vs. Scan-inserted netlist (Implementation). Scan test ports (`test_mode`, `SE`) were constrained to logic 0, and scan input/output ports (`SI[3:0]`, `SO[3:0]`) were marked as non-verify points because they exist only post-DFT.

3. **Post-PnR Verification (`post-PnR`):** Golden RTL (Reference) vs. Final exported PnR netlist `PnR/export/System_Top.v` (Implementation). SVF guidance loaded from `DFT/System_Top.svf`. Test controls were constrained to functional mode (`test_mode = 0`, `SE = 0`).

8.2 Verification Results Across All Stages

Table 12 presents the formal verification compare point statistics across all three stages.

Table 12: Formality Equivalence Checking Verification Results

Verification Stage	Passing Points	Failing	Aborted	Unverified	Verification Status
Post-Synthesis (`post-syn`)	361	0	0	0	**Verification SUCCEEDED**
DFF Registers	358	0	0	0	All flip-flops matched
Latches	1	0	0	0	Clock gating latch matched
Primary Ports	2	0	0	0	Unchanged primary ports
Post-DFT (`post-dft`)	362	0	0	0	**Verification SUCCEEDED**
Scan-replaced DFFs	359	0	0	0	Matched under test_mode=0
Latches	1	0	0	0	Clock gating latch matched
Primary Ports	2	0	0	0	Functional ports matched
Post-PnR (`post-PnR`)	362	0	0	0	**Verification SUCCEEDED**
Physical DFFs	359	0	0	0	Matched post-routing & buffer insertion
Physical Latches	1	0	0	0	Clock gating latch preserved
Primary Ports	2	0	0	0	Post-PnR ports matched

In all three runs, Formality reported **Verification SUCCEEDED** with zero failing, zero aborted, and zero unverified points, proving total mathematical fidelity from RTL source to final GDSII-prep netlist.

9 Gate-Level Simulation (GLS)

Gate-level simulation was performed in Mentor QuestaSim-64 (v10.7c) to validate functional and timing behavior on the final exported physical netlist (`PnR/export/System_Top.v`) using TSMC 130 nm standard cell models (`tsmc13_m.v`) with full SDF-max back-annotation (`PnR/export/System_Top.sdf`).

9.1 Simulation Configuration & Testbench Adaptation

The SystemVerilog testbench was invoked with the `+define+GLS` macro. The post-PnR netlist exposes physical scan test ports that are absent in the functional RTL; for functional GLS, these ports are tied off:

```
System_Top DUT (
    .REF_CLK(REF_CLK), .UART_CLK(UART_CLK), .RST_N(RST),
```

```
    .UART_RX_IN(RX_IN), .UART_TX_O(TX_OUT),
    .parity_error(parity_error), .framing_error(framing_error),
    // Tie off DFT test ports for functional GLS:
    .test_mode(1'b0),
    .SE(1'b0),
    .scan_clk(1'b0),
    .scan_rst(1'b0),
    .SI(4'b0),
    .SO() // Output floating
);
```

The simulator was launched with full timing check and delay transport options:

```
vsim -sdfmax /System_Top_tb/DUT=../PnR/export/System_Top.sdf -sdfnoerror \
     +transport_int_delays +neg_tchk -negdelay +no_notifier +DUMP +QUICK work.
         System_Top_tb
```

9.2 Timing Check Results & Functional Pass Rate

The simulation ran across **19.489 ms** of simulated real-time (covering start-up, configuration, register sweep, and all 15 ALU operations).

- **Timing Violations: Zero** setup, hold, recovery, removal, or pulse-width violation messages occurred throughout the entire 19.49 ms run.
- **Functional Tests:** All **61 functional assertions passed** with zero failures (`PASSED: 61, FAILED: 0`).
- **Signal Integrity:** Functional signals remained free of unknown (`X`) states or spurious transitions.

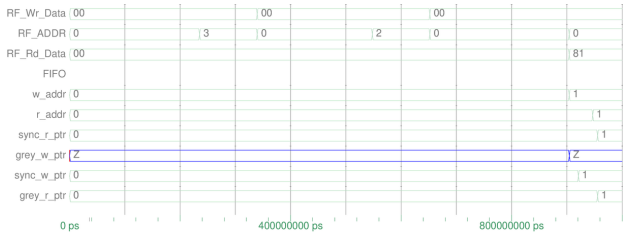
9.3 Waveform Snippet Analysis

The gate-level simulation waveforms (Figures 12 and 13) confirm accurate physical execution:

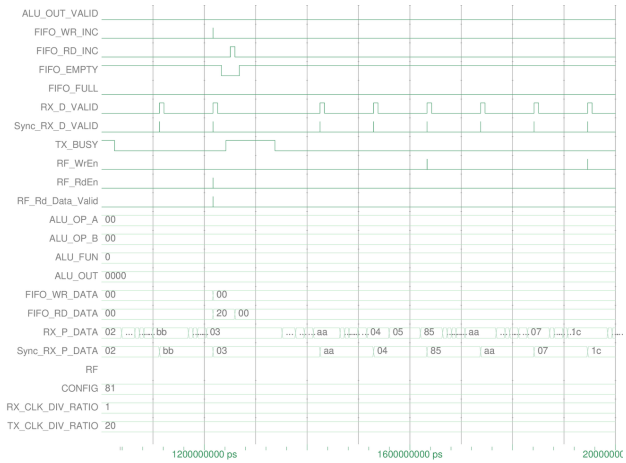
- **Snippet 1 (0–1.2 ms, Figure 12a):** Start-up register writes (`AA 03 20` and `AA 02 81`) pulse `RF_WrEn` without response. The first read (`BB 02`) pulses `RF_RdEn`, captures `0x81`, pushes to FIFO, and transmits on `TX_OUT`. Gray write pointer advances correctly.
- **Snippet 2 (1.2–2.4 ms, Figure 12b):** Read-back of REG3 returns `0x20` through the FIFO. Writes to registers 0x04 (`0x85`) and 0x07 (`0x1C`) execute with zero FIFO activity.
- **Snippet 3 (3.2–4.4 ms, Figure 12c):** Sequential reads of 0x07, 0x0B, and 0x0F return `0x1C`, `0xFF`, and `0x21`, matching previously written values.
- **Snippet 4 (4.4–4.7 ms, Figure 12d):** `CC C8 07 00` (ADD command). Operands A=`0xC8` and B=`0x07` are loaded; `ALU_OUT_VALID` pulses together with `FIFO_WR_INC`, pushing `0xCF` (`0xC8` + `0x07`).
- **Snippet 5 (15.2–16.4 ms, Figure 13):** Preload of REG0=`0x7A` and REG1=`0x0C` while the FIFO drains previous responses (`0x5D`, `0x00`, `0x74`).

10 Sign-Off Power Analysis (PrimeTime PX)

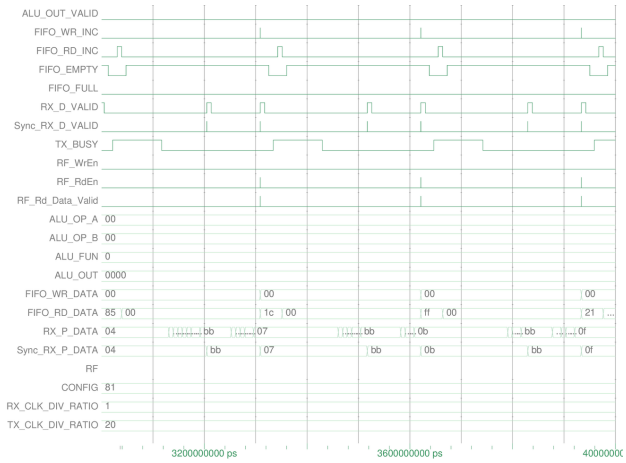
Sign-off power analysis was conducted in Synopsys PrimeTime PX (vO-2018.06-SP1) using time-based mode driven by switching activity extracted from gate-level simulation.



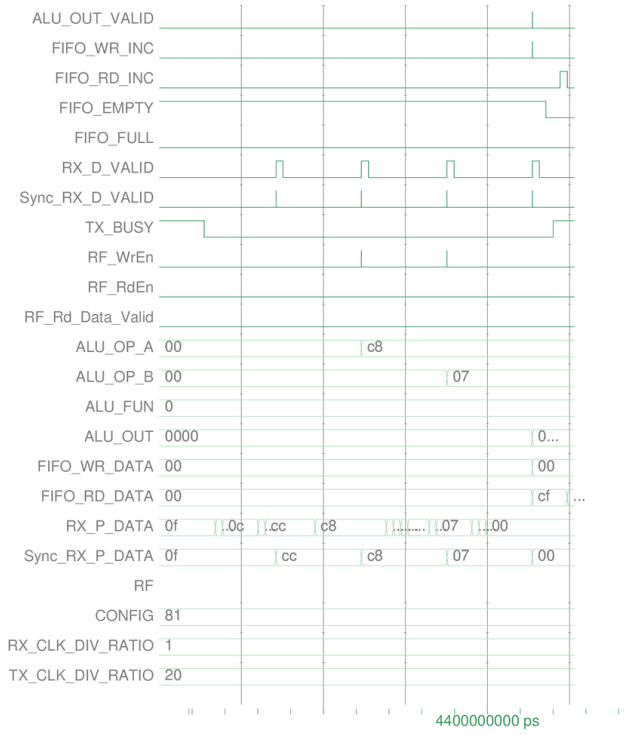
(a) Snippet 1: Start-up writes and first register read.



(b) Snippet 2: Read of REG3 and writes to 0x04/0x07.



(c) Snippet 3: Reads of registers 0x07, 0x0B, and 0x0F.



(d) Snippet 4: ALU operation with operands (`CC ADD`).

Figure 12: QuestaSim post-PnR gate-level simulation waveform snippets (SDF-max back-annotated).

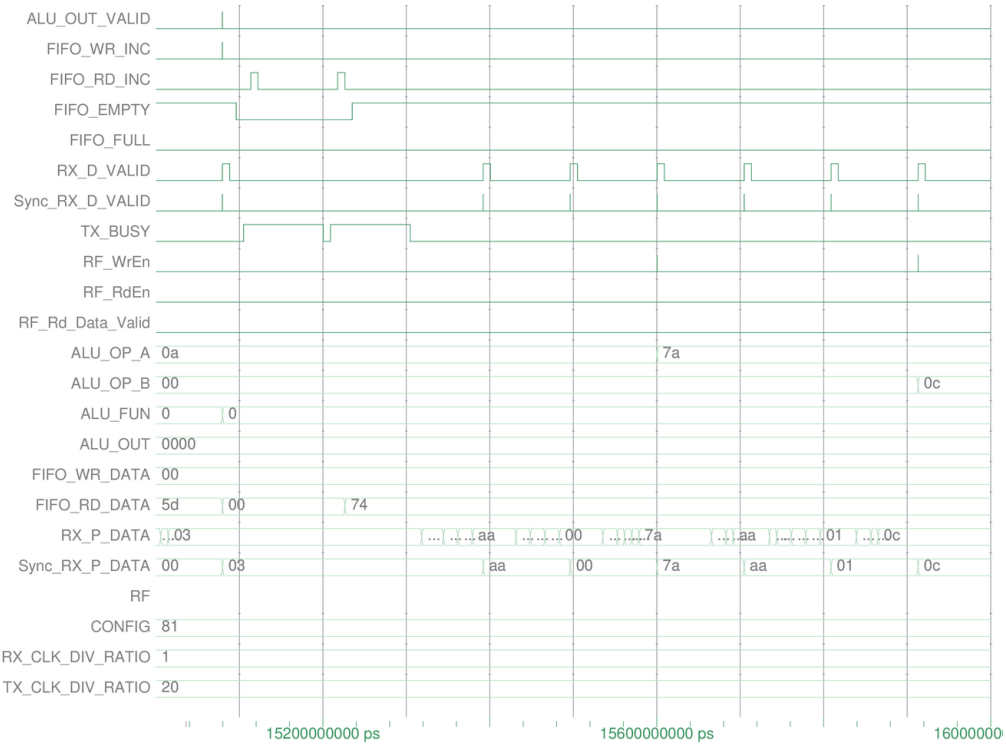


Figure 13: Snippet 5: Operand preload and FIFO pointer crossings in post-PnR gate-level simulation.

10.1 Power Analysis Methodology

Unlike early vectorless estimations that assume statistical toggle rates (e.g., 0.1 toggles/cycle), PrimeTime PX reads the actual circuit switching activity from the VCD file generated during gate-level simulation alongside extracted post-route RC parasitics:

- **Netlist, DSPF Parasitics & SDF:** The post-PnR netlist (`PnR/export/System_Top.v`) was linked with TSMC CL013G libraries and back-annotated with extracted post-route RC parasitics (`PnR/export/System_Top.spf` in DSPF format: 2,086 nets, 39,020 capacitances, and 36,934 resistances with 0 errors) and delays (`PnR/export/System_Top.sdf`: 16,960 cell delay arcs, 6,288 net arcs).
- **VCD Activity Annotation:** Switching activity was read using `read_vcd -strip_path System_Top_tb/D ../System_Top_tb.vcd`. **Annotation Coverage:** 2,096 out of 2,105 nets were fully annotated (**99.57% net annotation**), and 2,019 out of 2,019 leaf cells were fully annotated (**100.00% cell annotation**).
- **Analysis Mode:** Time-based dynamic analysis (`set_app_var power_analysis_mode time_based`) evaluated power across the full simulation window of 0 ns to 1.94887×10^7 ns (19.49 ms) at a 0.001 ns (1 ps) sampling interval.

10.2 Power Analysis Results

The verbose sign-off power report (`GLS/PT/report/System_PW.rpt`, reported in base units of 1 W) shows a total time-averaged power consumption of $\mathbf{8.459 \times 10^{-4}\,W}$ **(0.846 mW / 845.9 μW)**. The breakdown across power groups is detailed in Table 13.

10.3 Cross-Stage Power Comparison

Table 14 compares power estimates across the design flow, illustrating the fidelity gained through VCD-driven and DSPF-annotated sign-off.

Table 13: PrimeTime PX Sign-Off Power Breakdown (Time-Based VCD + DSPF Analysis)

Power Group	**Internal (μW)**	**Switching (μW)**	**Leakage (μW)**	**Total (μW)**	**Share (%)**
`clock_network` (i)	572.600	255.900	2.848	831.300	98.27%
`register`	0.331	0.128	6.119	6.578	0.78%
`combinational`	0.195	0.219	7.621	8.035	0.95%
`sequential`	0.000	0.000	0.000	0.000	0.00%
Total System Power	**573.100**	**256.200**	**16.590**	**845.900 μW**	**100.00%**
Distribution: Internal = 67.75% (573.1 μW), Switching = 30.29% (256.2 μW), Leakage = 1.96% (16.59 μW)					
Peak Instantaneous Power (1 ps bin): 0.1846 W at $t = 269,371.205$ ns (269.37 μs); Glitching = 0.0 W					

Table 14: Power Estimation Comparison Across Flow Stages

Design Stage	**Tool**	**Activity Source**	**Total Power**	**Analysis Characteristics**
Pre-DFT Synthesis	Design Compiler	Vectorless (0.1 toggle)	0.439 mW	Pre-CTS wireload estimate (no clock tree buffers)
Post-Route PnR	First Encounter	Default 20% PI toggle	1.556 mW	Post-CTS routed RC with active `scan_clk` (0.96 mW)
Sign-Off Analysis	PrimeTime PX	GLS VCD + DSPF RC	**0.846 mW**	**Sign-Off Golden: Real VCD switching + 16.59 μW leakage**

Notably, PrimeTime PX's cell leakage power (16.59 μW) matches Cadence Encounter's post-route leakage (16.62 μW) to within 0.2%, while total dynamic power drops from Encounter's 1.556 mW estimate to 0.846 mW because `scan_clk` is tied off in functional mode and the ALU clock is gated during idle UART intervals.

11 Conclusion and Sign-Off Matrix

The Low-Power Configurable Multi-Clock System has successfully completed all stages of the digital ASIC design flow, progressing from high-level Verilog/SystemVerilog RTL to verified GDSII layout. Table 15 summarizes the final sign-off status across all design and verification tools.

Table 15: Master Design Sign-Off Matrix

Flow Stage	Key Verification Metric	Result	Sign-Off Evaluation
RTL Design & Lint	SpyGlass Lint	0 errors / 0 warnings	**PASSED**
CDC Verification	SpyGlass CDC (39 crossings)	0 unsync / 0 glitches	**PASSED**
RDC Verification	SpyGlass RDC	0 RDC violations	**PASSED**
Logic Synthesis	DC Setup Slack / Hold Slack	+0.02 ns / +0.34 ns	**PASSED (All groups Met)**
DFT Scan Insertion	ATPG Stuck-At Test Coverage	**99.61%**	**PASSED (Target $\geq$ 95%)**
Formality Post-Syn	Formal Equivalence Check	361 compare points	**PASSED (Succeeded)**
Formality Post-DFT	Formal Equivalence Check	362 compare points	**PASSED (Succeeded)**
Formality Post-PnR	Formal Equivalence Check	362 compare points	**PASSED (Succeeded)**
Clock Tree Synth.	CTS Skew (`REF_CLK`)	109.3 ps	**PASSED (200 ps spec)**
Post-Route Timing	Setup WNS / Hold WNS	+0.364 ns / +0.074 ns	**PASSED (Zero DRVs)**
Design Rule Check	Encounter Geometry DRC	0 DRC violations	**PASSED**
Connectivity (LVS)	Encounter LVS	0 Open / 0 Short	**PASSED**
Antenna Verification	Process Antenna	0 violations	**PASSED**
Metal Density	Metal Density 20%–80%	0 violations	**PASSED**
Gate-Level Sim.	QuestaSim GLS w/ SDF-max	61/61 tests PASS	**PASSED (Zero violations)**
Sign-Off Power	PrimeTime PX (VCD + DSPF)	**0.846 mW**	**PASSED (845.9 μW avg)**

11.1 Key Technical Achievements

1. **Metastability-Proof Multi-Clock Domain Architecture:** Bridging 100 MHz `REF_CLK` and 3.6864 MHz `UART_CLK` with dual-clock Gray FIFOs, 2-stage multi-bit bus synchronizers, and reset synchronizers, validated clean with zero SpyGlass CDC/RDC violations.

2. **High-Coverage Manufacturing Testability:** 4 scan chains inserted with full functional clock and reset multiplexing, delivering 99.61% stuck-at test coverage (15,505 detected faults).

3. **Physical Design Closure with Zero Violations:** Full physical routing on 6 metal layers with clean CTS, zero DRVs, positive setup and hold slacks, and 100% clean DRC/LVS/Antenna/Density sign-off.

4. **Proven End-to-End Silicon Fidelity:** Formality equivalence succeeded across all 3 flow transformations, complemented by 61/61 passing gate-level simulation vectors under maximum SDF delay annotation.

5. **Low-Power Gated Clock Operation:** PrimeTime PX sign-off power analysis with extracted DSPF parasitics and GLS switching activity proved a total average dissipation of 0.846 mW

(845.9 μW), demonstrating the effectiveness of the integrated clock gating and multi-rate divider architecture.